EFRAIN RIVAS

● ENTERPRISE CLOUD & KUBERNETES ARCHITECT

SENIOR CLOUD ARCHITECT & INFRASTRUCTURE SYSTEMS ENGINEER

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// 01 PROFESSIONAL PROFILE & ARCHITECTURAL FOCUS

Results-driven **Senior Cloud Architect and Systems Engineer** with 20+ years of hands-on enterprise infrastructure experience designing, modernizing, and automating mission-critical systems across Microsoft Azure, Kubernetes, Active Directory Domain Services, Linux, and multi-hypervisor virtualization clusters. Proven architectural track record deploying global multi-site Active Directory forests across 7 global data centers, engineering 5-stage financial CI/CD release pipelines with 99.999% SLA uptime, orchestrating high-availability containerized microservices clusters, and implementing zero-touch endpoint provisioning via Microsoft Intune and Windows Autopilot. Creator of high-performance open-source platforms and engineering tools (**NovaFlix, AegisDNS, Pulsearr, Renamarr, Proxmox Cluster Suite, 58+ Intune remediation modules**) with deep expertise in PowerShell, Python, Proxmox VE clustering, and enterprise identity governance.

// 02 CORE TECHNICAL COMPETENCIES & SYSTEMS MATRIX

▶ CLOUD & DIRECTORY ARCHITECTURE

Microsoft Azure | Entra ID (Azure AD) | Active Directory DS | Hybrid Identity (PHS/PTA) | VNets / NSGs | Azure Files | Group Policy (GPO) | Exchange Online | M365 Tenant Admin

▶ CONTAINERS & KUBERNETES

Kubernetes (k3s / k8s) | Control-Plane HA | Traefik Ingress | Worker Pools | NFS Persistent Volumes | Pod Auto-Recovery | Rolling Deployments | Docker Containers

▶ VIRTUALIZATION & HYPERVISORS

Proxmox VE (HA Cluster) | VMware vSphere / ESXi | Linux KVM | Citrix XenServer | Microsoft Hyper-V | Zero-Balloon Memory Tuning | Azure Virtual Desktop (AVD) | FSLogix Profiles | SAN / NFS Clustering

▶ AUTOMATION & DEVOPS ENGINEERING

PowerShell 7+ | Python (Asyncio/FastAPI) | Bash | CI/CD Release Pipelines | Azure Runbooks | Systemd Service/Timers | Git / GitHub Actions | REST APIs & WebSockets

▶ ENDPOINT & DEVICE GOVERNANCE

Microsoft Intune (MDM/MAM) | Windows Autopilot | SCCM / MECM | Win32 Packaging (.intunewin) | Proactive Remediations | BitLocker Key Escrow | Patch Compliance Governance | Windows 10/11 & macOS

▶ ZERO TRUST, DNS & SECURITY

Zero Trust Security | Automated SSL (DNS-01) | Pi-hole v6 / Unbound DNSSEC | Privileged Identity (PIM) | Conditional Access | SAML 2.0 / OAuth SSO | FIDO2 / MFA | SecOps Hardening | Defender for Cloud

// 03 ENTERPRISE INFRASTRUCTURE EXPERIENCE

Senior IT Infrastructure Consultant ● O.S. Consulting, Inc. — Remote | 2025 – Present

- **Enterprise Cloud & Systems Advisory**: Deliver senior-level infrastructure consulting across diverse enterprise client environments, designing scalable cloud architectures, virtual server platforms, and network topologies tailored to compliance and business requirements.
- **High-Availability Kubernetes Orchestration**: Architect and deploy resilient multi-node Kubernetes clusters with control-plane replication, worker pools, Traefik ingress routing, and shared NFS persistent storage; engineered 4-replica zero-downtime deployments with health checks and automated failover.
- **Proxmox VE Fleet Optimization**: Administer multi-node Proxmox VE virtualization clusters, tuning hardware resource allocation, eliminating memory ballooning contention for guaranteed performance, and orchestrating live-migration and automated backup routines.
- **DNSSEC & Automated SSL/TLS Lifecycle**: Engineered tri-node synchronized Pi-hole v6 and Unbound recursive DNS clusters with root-hint recursion and DNSSEC validation; built automated Let's Encrypt wildcard certificate renewal pipelines via Cloudflare DNS-01 APIs and systemd timers.
- **Hybrid Identity & Modern Endpoint**: Architect hybrid AD to Entra ID synchronizations with PHS and Conditional Access; modernize SCCM/GPO environments to Intune with zero-touch Autopilot.
- **Tier 3 Root Cause Analysis (RCA)**: Act as the principal escalation architect for complex hypervisor outages, directory replication failures, and network latency anomalies, authoring automated remediation runbooks to eliminate systemic recurring incidents.

Senior Systems Administrator & Infrastructure Architect • BlueMatrix

Durham, NC | 2018 – 2025

- **Global Multi-Site Active Directory Forest**: Architected, built, and maintained enterprise Active Directory forest and domain infrastructure across 7 global data centers and regional offices; tuned multi-site replication topology, site-cost weighting, and distributed DNS to guarantee sub-second authentication worldwide.
- **Multi-Hypervisor Virtualization Clusters**: Administered and tuned high-density virtualization clusters across Linux KVM, Citrix XenServer, VMware vSphere/ESXi, and XCP-NG; orchestrated shared SAN/NFS arrays, vCPU scheduling, dynamic memory allocations, and automated snapshot replication.
- **Azure Cloud Infrastructure & Automation**: Spearheaded phased migrations transitioning mission-critical on-premises systems to Microsoft Azure; architected VNets, Network Security Groups (NSGs), Azure Virtual Machines, and custom Azure Automation Runbooks to govern automated compute scaling and patch compliance.
- **Automated Identity Lifecycle Engineering**: Developed robust PowerShell automation workflows connecting HR systems with AD DS and Entra ID, completely automating provisioning, RBAC group delegation, Microsoft 365 licensing, and zero-day offboarding security revocation.

DevOps Engineer & Systems Administrator • StreamingEdge (Tradition America)

New York, NY (Hybrid) | 2013 – 2018

- **Financial CI/CD Release Pipeline Engineering**: Directed release engineering across a structured 5–stage deployment pipeline (DEV ➔ QA ➔ UAT ➔ Staging ➔ Production) for institutional financial trading systems, coordinating zero-downtime off-hours application releases and configuration verification.
- **Mission-Critical Systems Reliability (99.999% SLA)**: Engineered and administered high-availability Windows Server (2012–2022) and Linux (RHEL, CentOS, Ubuntu) server farms hosting high-throughput financial transaction engines, consistently maintaining five-nines (99.999%) SLA uptime.
- **Clustered Enterprise Virtualization**: Managed fault-tolerant virtualization clusters across Microsoft Hyper-V and VMware vSphere, configuring multi-NIC teaming, virtual switch isolation, host failover clustering, and disaster recovery replication.
- **Enterprise Networking & Perimeter Security**: Designed and maintained high-throughput LAN/WAN routing, core switching, hardware firewalls, site-to-site IPsec VPN tunnels, and enterprise telecommunications infrastructure.

Senior Network Administrator • Digital Network Development

Peoria, IL | 1999 – 2011

- **Enterprise Windows Server Operations**: Commissioned, upgraded, and administered physical enterprise servers spanning Windows NT 4.0, 2000, 2003, and 2008, managing Active Directory, Group Policy Objects, DNS, DHCP, and enterprise storage.
- **Telecommunications & Core Networking**: Engineered, programmed, and maintained managed Cisco switches, routers, multi-building wireless links, T1/T3 telecommunications circuits, and enterprise VOIP telephony platforms.
- **Disaster Recovery & Tier 3 Escalation**: Managed enterprise tape and NAS backup routines, storage arrays, OS bare-metal imaging (WDS/MDT), and routine patch compliance across distributed server farms.

// 04 FLAGSHIP ENGINEERING PROJECTS & OPEN SOURCE TOOLS

NovaFlix (Creator & Architect) `github.com/digdast/novaflix`
Sovereign open media platform & home theater engine (Plex/Jellyfin alternative) with 4K HDR Direct Play, HTTP 206 Range streaming, built-in remote access, and SyncPlay.

AegisDNS (Creator & Architect) `github.com/digdast/aegisdns`
Recursive DNS resolver & RFC 8484 DNS-over-HTTPS (DoH) engine traversing root hints with zero telemetry, 227k+ threat sinkhole rules, and real-time cyber UI.

HA Kubernetes Cluster & Ingress `resume.ertechlab.com`
Orchestrated multi-node k8s cluster with Traefik ingress, shared NFS storage, and automated 4–replica zero-downtime web application hosting.

Proxmox Cluster Suite `github.com/digdast/proxmox-cluster-suite`
Automated enterprise optimization, no-sub repo activation, nag removal, CPU governor, memory ballooning elimination, and real-time HA monitoring.

Pulsearr (Creator & Architect) `github.com/digdast/pulsearr`
Architected an asynchronous Python media suite unifying Sonarr, Radarr, and Prowlarr into a single lightweight daemon; built with FastAPI, asyncio, and SQLite WAL mode (85MB idle RAM).

Enterprise Intune Automation Catalog `github.com/digdast/Intune-scripts`
Authored and open-sourced 58 production PowerShell automation modules covering Autopilot pre-provisioning, BitLocker cipher detection, and proactive endpoint remediations.

// 05 INDUSTRY CERTIFICATIONS & TECHNICAL CREDENTIALS

✓ Microsoft Certified: Azure Administrator Associate (AZ-104)
✓ Microsoft 365 Certified: Administrator Expert (MS-102)
✓ Microsoft Entra ID: Administration & Identity Security
✓ CompTIA Network+ Certified
✓ CompTIA Security+ Certified
✓ Inter-Tel University: Telecommunications & VOIP Systems